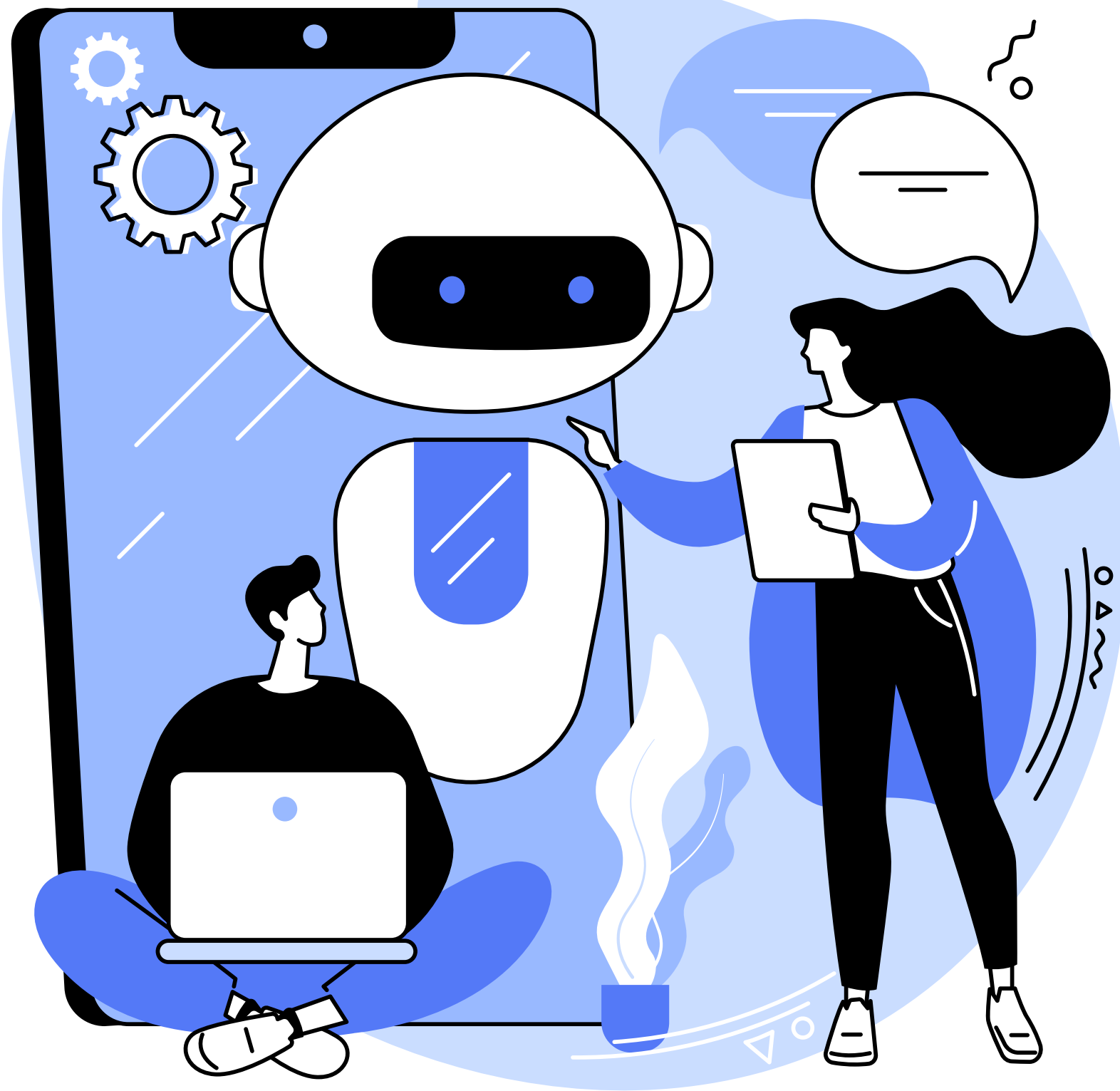




MOBILENET

V1-V2-V3 IMPLEMENTATION ON CPU

Pre-trained models used to measure speed and accuracy of models on CPU

```
# Load models  
mobilenet_v1 = timm.create_model('mobilenetv1_100', pretrained=True).to(device).eval()  
mobilenet_v2 = models.mobilenet_v2(pretrained=True).to(device).eval()  
mobilenet_v3 = models.mobilenet_v3_large(pretrained=True).to(device).eval()
```

All models are pretrained on ImageNet-1k

```
def benchmark_model(model, batch_size, runs):
    dummy_input = torch.randn(batch_size, 3, 224, 224).to(device)
    times = []
    with torch.no_grad():
        for _ in range(runs):
            start = time.time()
            _ = model(dummy_input)
            end = time.time()
            times.append(end - start)
    avg_time = sum(times) / len(times)
    throughput = batch_size / avg_time
    return throughput
```

Generate a random batch of 8 images with resolution 224×224 (standard for ImageNet).

**For each run:
Record start and end time.
Measure how long inference takes.**

$$\text{Throughput (images/sec)} = \frac{\text{batch size}}{\text{average inference time}}$$

🔧 THREADS: 1
MobileNet V1: 50.19 images/sec
MobileNet V2: 41.80 images/sec
MobileNet V3: 42.04 images/sec

🔧 THREADS: 2
MobileNet V1: 77.67 images/sec
MobileNet V2: 69.08 images/sec
MobileNet V3: 65.45 images/sec

🔧 THREADS: 3
MobileNet V1: 93.98 images/sec
MobileNet V2: 79.99 images/sec
MobileNet V3: 72.43 images/sec

🔧 THREADS: 4
MobileNet V1: 93.80 images/sec
MobileNet V2: 81.61 images/sec
MobileNet V3: 76.31 images/sec

MobileNet V1 is the Fastest
It's the lightest model, with the simplest architecture.

Achieves the highest throughput in all thread settings.

MobileNet V2 and V3 Trade Speed for Accuracy

These make them more accurate, but also slightly slower on CPU.

🔧 THREADS: 1
MobileNet V1: 50.19 images/sec
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MobileNet V3: 42.04 images/sec

🔧 THREADS: 2
MobileNet V1: 77.67 images/sec
MobileNet V2: 69.08 images/sec
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MobileNet V1: 93.80 images/sec
MobileNet V2: 81.61 images/sec
MobileNet V3: 76.31 images/sec

**Multithreading Improves Performance
— But Only Up to a Point**

**From 1 → 2 → 3 threads, throughput
increases significantly.**

At 4 threads, performance saturates:

**Likely because the CPU hits its
memory/cache bandwidth limit**

**Or physical core limit (many
laptops/desktops have 4–6 cores)**

**Going beyond 4 threads adds thread
overhead and less efficiency**

🔧 THREADS: 1
MobileNet V1: 50.19 images/sec
MobileNet V2: 41.80 images/sec
MobileNet V3: 42.04 images/sec

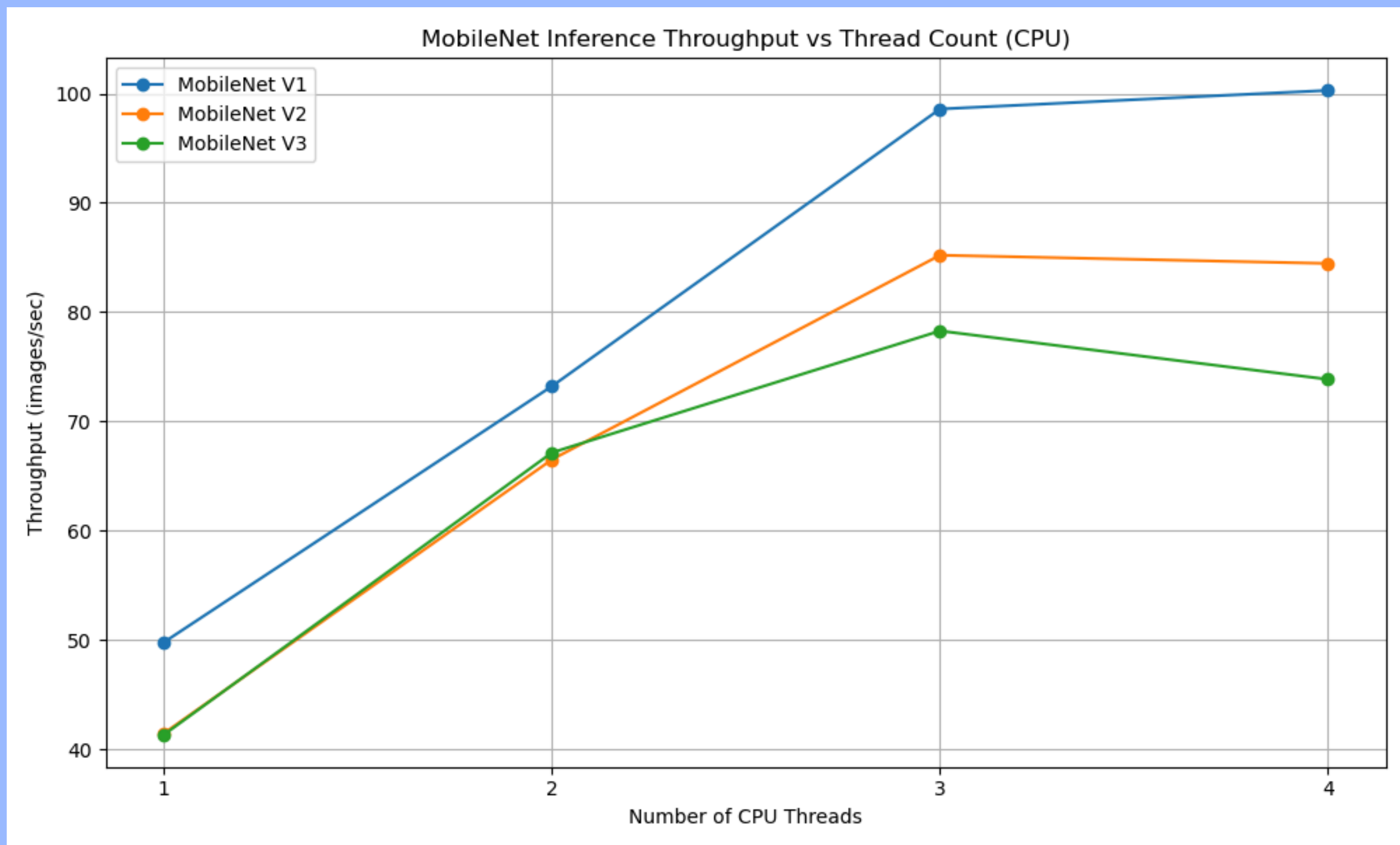
🔧 THREADS: 2
MobileNet V1: 77.67 images/sec
MobileNet V2: 69.08 images/sec
MobileNet V3: 65.45 images/sec

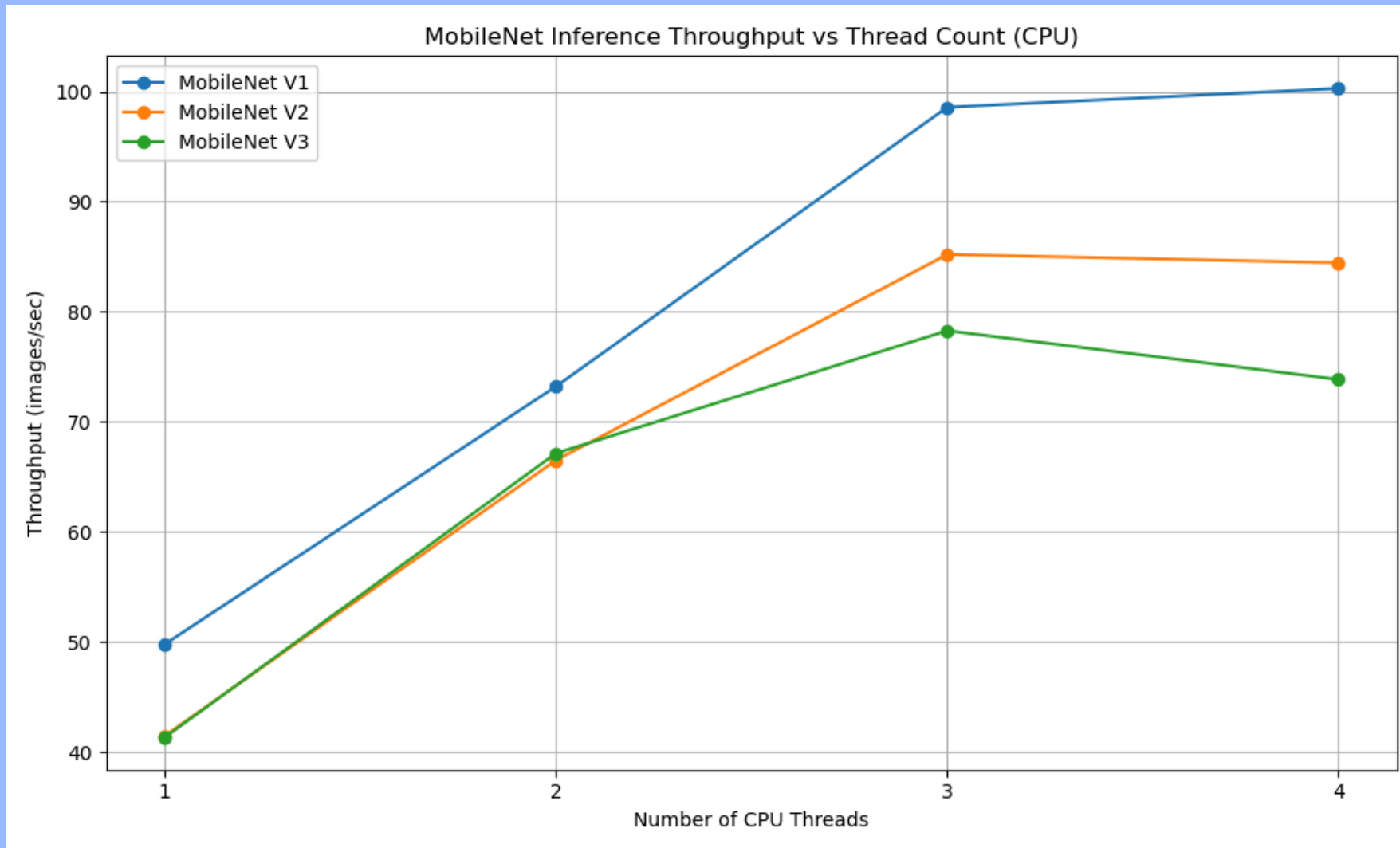
🔧 THREADS: 3
MobileNet V1: 93.98 images/sec
MobileNet V2: 79.99 images/sec
MobileNet V3: 72.43 images/sec

🔧 THREADS: 4
MobileNet V1: 93.80 images/sec
MobileNet V2: 81.61 images/sec
MobileNet V3: 76.31 images/sec

V1 Scales Better Than V3
V1 sees ~87% speedup from 1
→ 2 → 3 threads.

V3 sees ~50–60%
improvement — more
compute-heavy layers
struggle to scale.

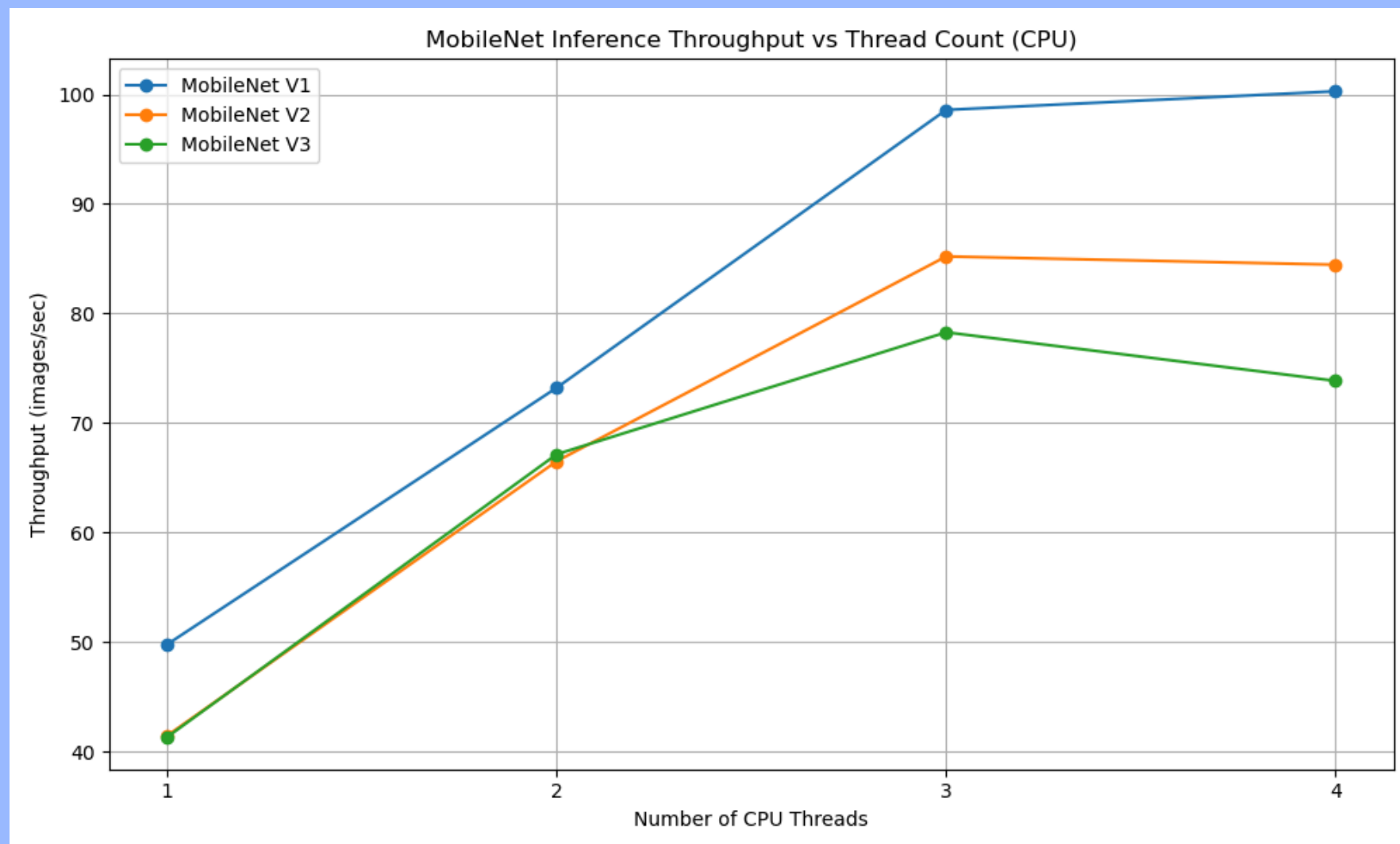




MobileNet V1 consistently delivers the highest throughput. Its simpler architecture makes it lightweight and highly parallelizable.

MobileNet V2 improves on V1 with slightly more complexity but maintains strong throughput. It scales well up to 3 threads, but gains taper at 4.

MobileNet V3 is the most accurate but also the slowest. Performance gains flatten and slightly drop at 4 threads, likely due to CPU memory/cache bottlenecks or thread contention.



Why Thread 4 Isn't Always Best

- Threading scales only to the extent your CPU's architecture allows.
- After ~3 threads, memory bandwidth, cache access, and thread overhead start to dominate.
- For small models like MobileNet, sweet spot = 3–4 threads on most CPUs.

```
100%|████████████████████████████████████████████████████████████████████████████████| 491/491 [00:43<00:00, 11.2  
1it/s]  
MobileNet V1 – Top-1: 9.07% | Top-5: 9.58%  
100%|████████████████████████████████████████████████████████████████████████████████| 491/491 [00:50<00:00, 9.7  
8it/s]  
MobileNet V2 – Top-1: 9.17% | Top-5: 9.71%  
100%|████████████████████████████████████████████████████████████████████████████████| 491/491 [00:59<00:00, 8.3  
0it/s]  
MobileNet V3 – Top-1: 9.43% | Top-5: 9.81%
```

Why We're Getting ~9% Accuracy

Evaluating the models on the imagenette2-160 dataset,
which:

Contains only 10 classes (a subset of ImageNet's 1,000
classes)

But pretrained models trained on 1,000-class full ImageNet

WHAT'S NEXT?

Try these models on GPU and see the results, compare them by CPU.

QUESTION

Try to label test images correctly to see expected accuracies of
MobileNet 70-80% of accuracies.